Profile Sieve EL with Nuisance Breaks

A Geometry-First Proof Map

One-sentence thesis. Break-adaptive projection is the decontamination step that turns false predictability from structural breaks into valid empirical-likelihood inference for β.

1. Gap: structural breaks can look like predictability

The predictive model is

$$Y_t = \beta_0 X_{t-1} + \mu_{0,t} + U_t, \qquad \mu_{0,t} = m_j(W_{t-1}) \quad \text{on regime } j.$$

The difficulty is not only that $\mu_{0,t}$ is nonparametric. It may also break. If inference residualizes with a stable/no-break nuisance space, its residual-maker M_{st} generally leaves a nonzero nuisance residue:

$$M_{\mathrm{st}}\mu_0 \neq 0.$$

At the true value β_0, the stable projected score is

$$S_T^{\mathrm{st}}(\beta_0) = T^{-1/2}u' M_{\mathrm{st}} w + T^{-1/2}\mu_0' M_{\mathrm{st}} w.$$

The second term is omitted-break drift:

$$B_T = T^{-1/2}\mu_0' M_{\mathrm{st}} w = \sqrt{T}\,\langle M_{\mathrm{st}}\mu_0, M_{\mathrm{st}} w\rangle_T.$$

Thus false predictability is a geometric angle problem. If the unremoved nuisance residue $M_{\mathrm{st}}\mu_0$ is not orthogonal to the residualized score direction $M_{\mathrm{st}} w$, the score can be large even when $\beta = \beta_0$.

Gap. Existing stable-nuisance predictability procedures can confuse nuisance structural breaks with predictive signal. The missing piece is a method that ties break selection, nuisance removal, and EL inference together so that break residue cannot enter the predictive score.

New idea. Treat a candidate break partition as a nuisance subspace:

$$\Lambda \longmapsto \mathcal{N}_{K,\Lambda,T} = \mathrm{col}(Q_{K,\Lambda}) \subset L^2(P_T), \qquad M_{K,\Lambda} = I - \Pi_{K,\Lambda,T}.$$

Break search is then subspace selection, not just date estimation.

Why it matters. Under the correct break-aware subspace,

$$M_0\mu_0 \approx 0,$$

whereas under a stable nuisance subspace $M_{\mathrm{st}}\mu_0 \not\approx 0$. The geometry identifies exactly which component must be removed before testing predictability.

2. Geometry: break search and inference happen in normal space

Let $L^2(P_T) = (\mathbb{R}^T, \langle a, b\rangle_T = T^{-1}a'b)$. For a trial β, write $y_\beta = y - \beta x$. Profiling over a block sieve nuisance is orthogonal projection:

$$RSS_K(\beta, \Lambda) = T\,\mathrm{dist}_T^2(y - \beta x, \mathcal{N}_{K,\Lambda,T}) = T\|M_{K,\Lambda}(y - \beta x)\|_T^2.$$

So the break estimator chooses the nuisance subspace closest to y_β. The projected score uses only normal-space components:

$$\Psi_T(\beta, \Lambda) = \langle M_{K,\Lambda}(y - \beta x), M_{K,\Lambda}w\rangle_T.$$

This is the decontaminated predictability moment.

Population identification. Let M_0^P project away the true break-aware nuisance tangent space. Since μ_0 belongs to that nuisance space,

$$M_0^P\mu_0 = 0.$$

The population moment satisfies

$$\Psi(\beta, \Lambda_0) = \langle M_0^P(Y - \beta X), M_0^P w\rangle_P = -(\beta - \beta_0)\Gamma, \qquad \Gamma = \langle M_0^P X, M_0^P w\rangle_P \neq 0.$$

Thus β_0 is identified by transversality: after quotienting out the nuisance space, the residualized parametric direction is not orthogonal to the chosen score direction.

Good-event proof map. Probability builds the random geometry. Once the following good event holds, the proof is deterministic:

$$\mathcal{G}_T = G_T^{iso} \cap G_T^{approx} \cap G_T^{angle} \cap G_T^{local} \cap G_T^{select} \cap G_T^{convex}.$$

Component	Meaning and use
G_T^{iso}	Gram and inner-product stability for projection and variance.
G_T^{approx}	$M_0\mu_0$ is small, so nuisance residue is negligible.
G_T^{angle}	$\Gamma \neq 0$ and score variance is nondegenerate.
G_T^{local}	$\widehat{\Lambda}$ replaces Λ_0 in score and variance.
G_T^{select}	RSS localizes breaks and selects the order.
G_T^{convex}	EL convex hull and max-score conditions hold.

The full stochastic proof verifies $\Pr(\mathcal{G}_T) \to 1$ through Gram laws, sieve approximation, RSS concentration, break replacement, CLT, variance LLN, max-score, and convex-hull bounds.

3. EL inference and the payoff

For fixed (β, Λ), define

$$g_t(\beta, \Lambda) = \{M_{K,\Lambda}(y - \beta x)\}_t \{M_{K,\Lambda} w\}_t.$$

The scalar empirical likelihood is a convex simplex problem:

$$\max_{\pi \in \Delta_T} \sum_{t=1}^{T} \log(T\pi_t) \quad \text{s.t.} \quad \sum_{t=1}^{T} \pi_t g_t(\beta, \Lambda) = 0.$$

Its KKT form is

$$\pi_t(\lambda) = \frac{1}{T\{1 + \lambda g_t\}}, \qquad P_T \left\{ \frac{g_t}{1 + \lambda g_t} \right\} = 0.$$

On $\mathcal{G}_T$, the logarithmic barrier has the local quadratic expansion

$$-2 \log EL(\beta_0, \widehat{\Lambda}) = \frac{T\{\Psi_T(\beta_0, \widehat{\Lambda})\}^2}{V_T(\widehat{\Lambda})} + o_p(1).$$

The break-aware projection bridge gives the oracle score reduction

$$\sqrt{T}\, \Psi_T(\beta_0, \widehat{\Lambda}) = T^{-1/2} u' M_0 w + o_p(1).$$

If

$$T^{-1/2} u' M_0 w \Rightarrow N(0, V), \qquad V_T(\widehat{\Lambda}) \to_p V > 0,$$

then

$$-2 \log EL(\beta_0, \widehat{\Lambda}) \Rightarrow \chi^2_1.$$

Why this fills the gap. The method turns unknown nuisance breaks into a subspace-selection problem:

$$\text{break partition} \Rightarrow \text{nuisance subspace} \Rightarrow \text{orthogonal residual} \Rightarrow \text{projected score} \Rightarrow \text{EL KKT} \Rightarrow \chi^2_1.$$

The contaminated stable score is replaced by the oracle break-aware score. That is why the geometry is central to the paper, not a notation choice.

Efficiency add-on. Under a conditional Gaussian location efficiency calculation with variance $\sigma^2(S, W)$, choose

$$w^* = \frac{X}{\sigma^2(S, W)}.$$

Then

$$M_0^P w^* = \frac{X - E(X \mid S, W)}{\sigma^2(S, W)},$$

so the projected score equals the efficient score. In the homoskedastic case, using $w = X$ gives the same root.